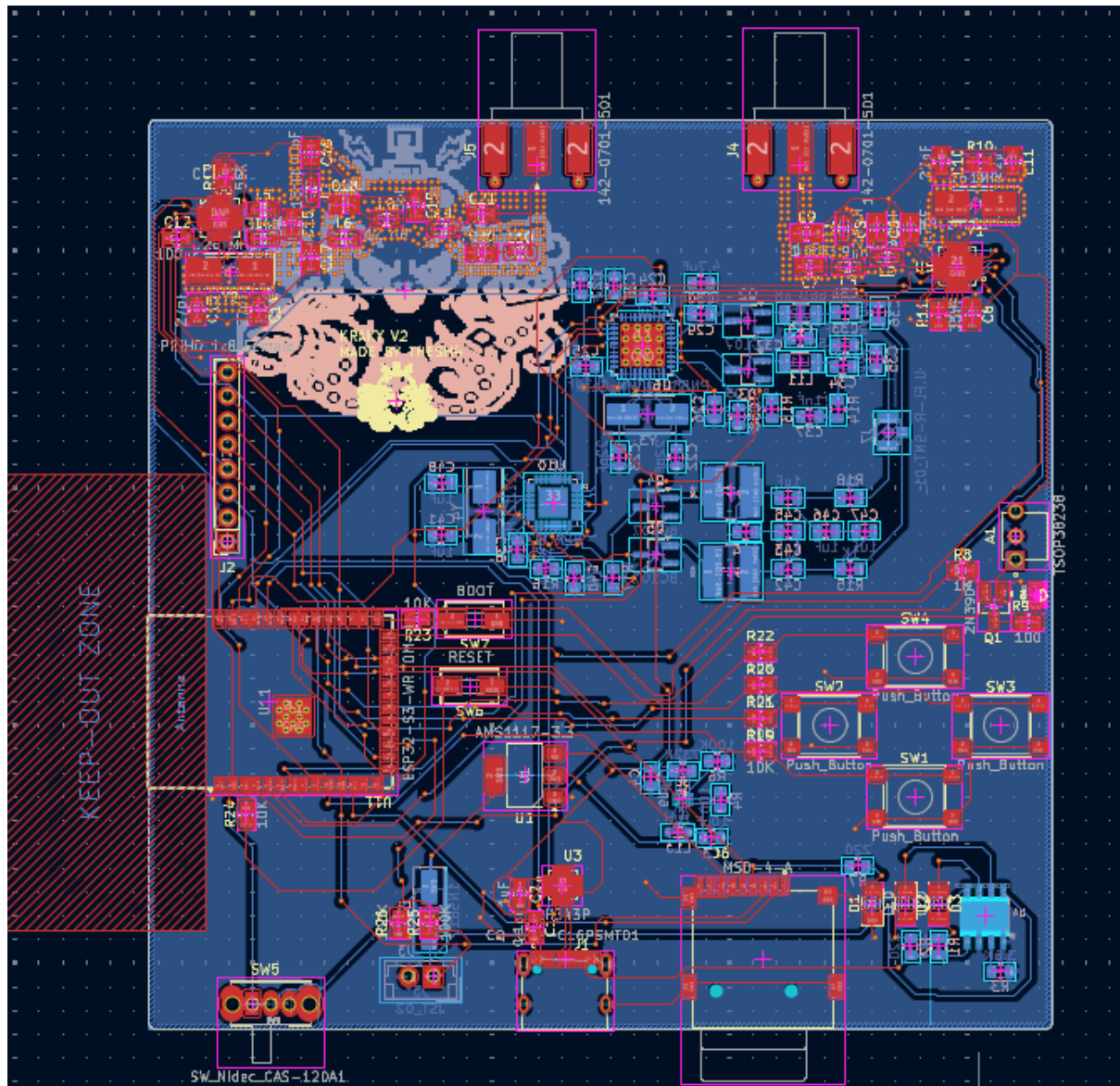


Kraky2 user manual

This is kraky2, a pentesting tool that is designed to analyze a lot of different radio frequencies, like wifi, bluetooth, sub-GHz and many more.

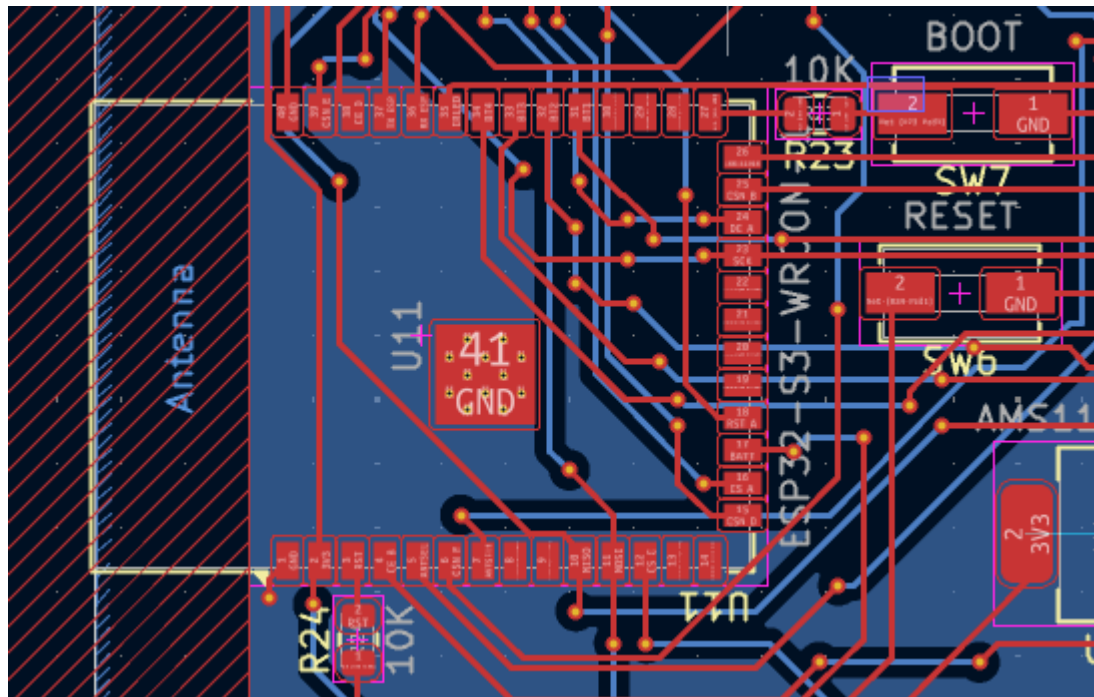


ESP32-S3

We have chosen the ESP32-S3 for the brain of this tool, because of his great capabilities and the possibility to introduce AI in this project later.

The ESP32-S3 handles all the communications between the modules, keeps the firmware and is responsible for the functionalities of the bluetooth.

Near the ESP32-S3 there are two buttons that have the function of boot and reset, so the it can be programmable.

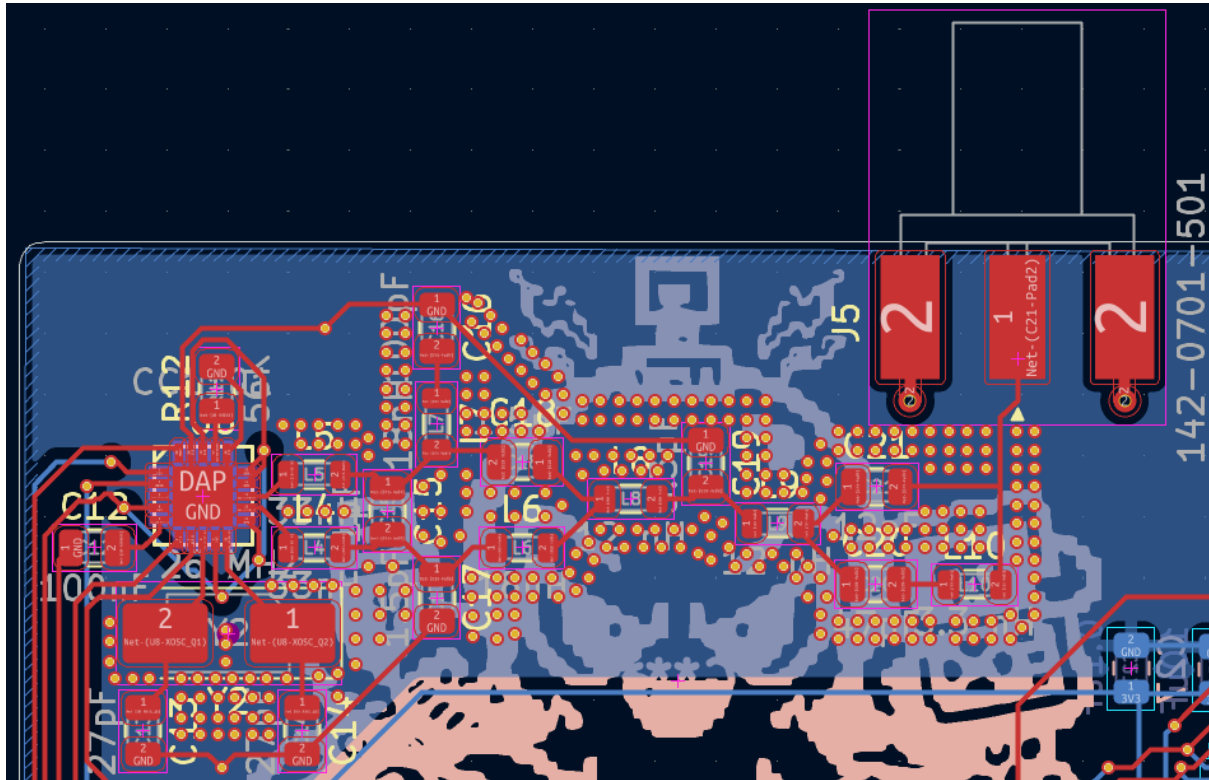


Sub-GHz

The sub-GHz function is handled by the CC1101 chip that is built to do 819MHz, this chip is one of the most capable in the sub-GHz category and is very convenient because it uses SPI, which if you have a lot of modules comes in very handy.

We built the antenna so it can have very little interference on the 819 MHz range, to build it we referenced the CC1101 manual.

The CC1101 is connected to the ESP32 with all the required wires for SPI and the CSN pin is connected to the GPIO pin number 3, and the CE pin is connected to the GPIO pin number 2.

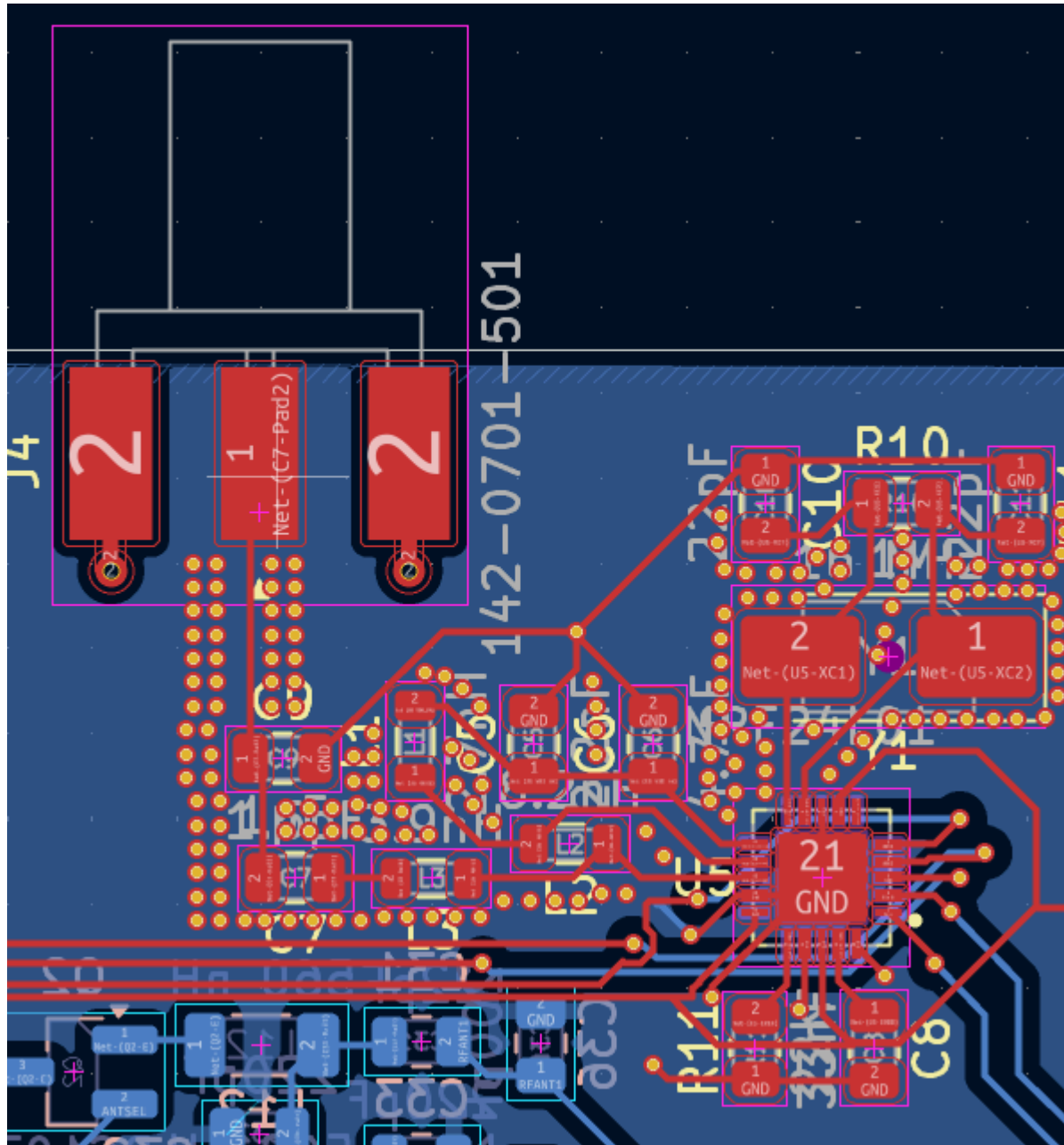


WIFI

The wifi function is handled by the NRF24L01 chip that is built to scan and monitor all the wifi signals that are in range, it is also connected via SPI.

We, like the CC1101, used the NRF24L01 manual to build the antenna.

The NRF24L01 is connected to the ESP32 with all the required wires for SPI and the CSN pin is connected to the GPIO pin number 48, and the CE pin is connected to the GPIO pin number 4.

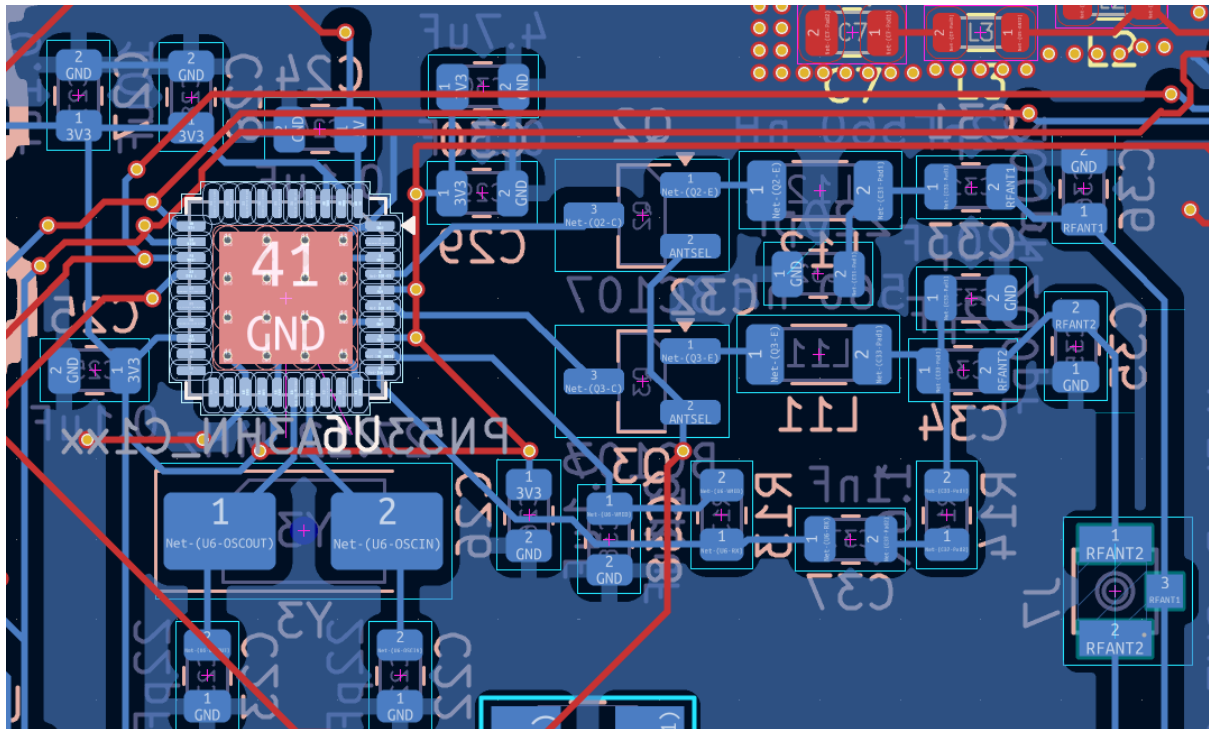


NFC

the NFC module is the PN532 and is situated on the other side of the board and serves the function of reading and writing on NFC tag, at the moment the only thing that can write are hello NFC and something custom that we will write but hopefully in the future you can write what you want on the NFC tags.

The PN532 is also connected in SPI and the CSN is the GPIO pin number 1.

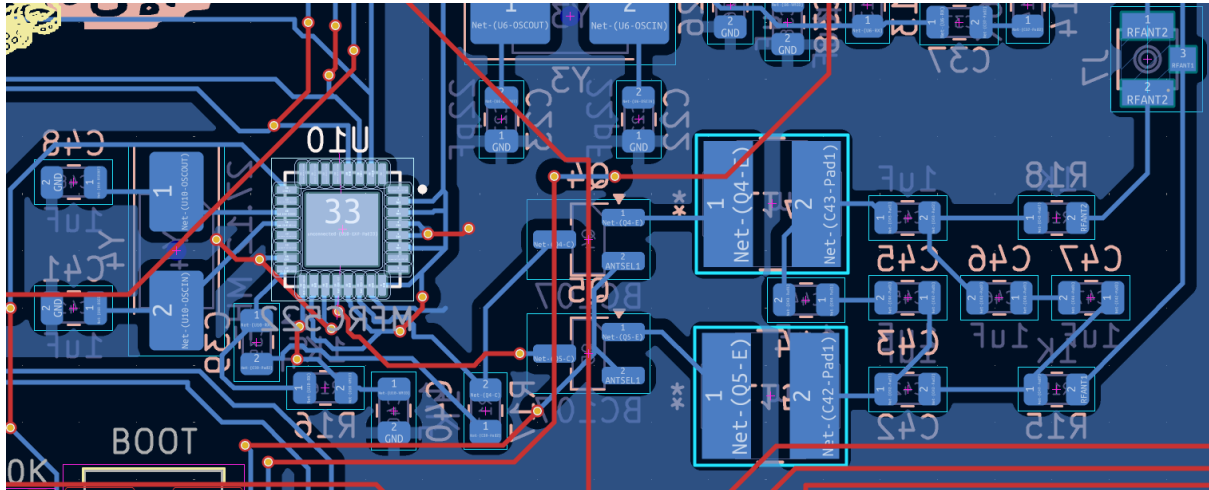
The NFC and RFID modules are connected together on the same antenna so there are 2 pairs of transistors that the NFC one is connected to the GPIO pin number 5.



RFID

The RFID module uses the MFRC522 chip and like the NFC module is situated on the other side of the board, for the development of the firmware we are a bit stuck because we don't own any MFRC522 modules or chips so we can't develop it at the moment.

The MFRC522 is connected with SPI and the CSN is the GPIO pin number 6, and the antenna selection pin is the GPIO pin number 7.

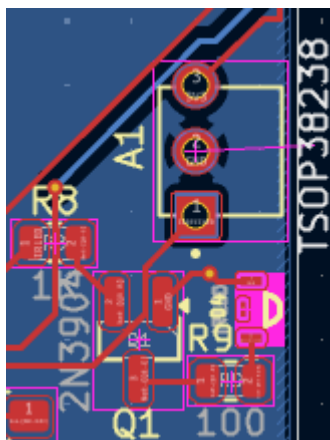


IR

The IR module is composed of an IR led connected to 2 resistors and a transistor so that the current that flows through it is limited, and a TSOP38238 that receives the IR waves and reads them.

The IR firmware is a bit complicated mainly because we don't have many things that we can try it on so it is a bit untested at the moment.

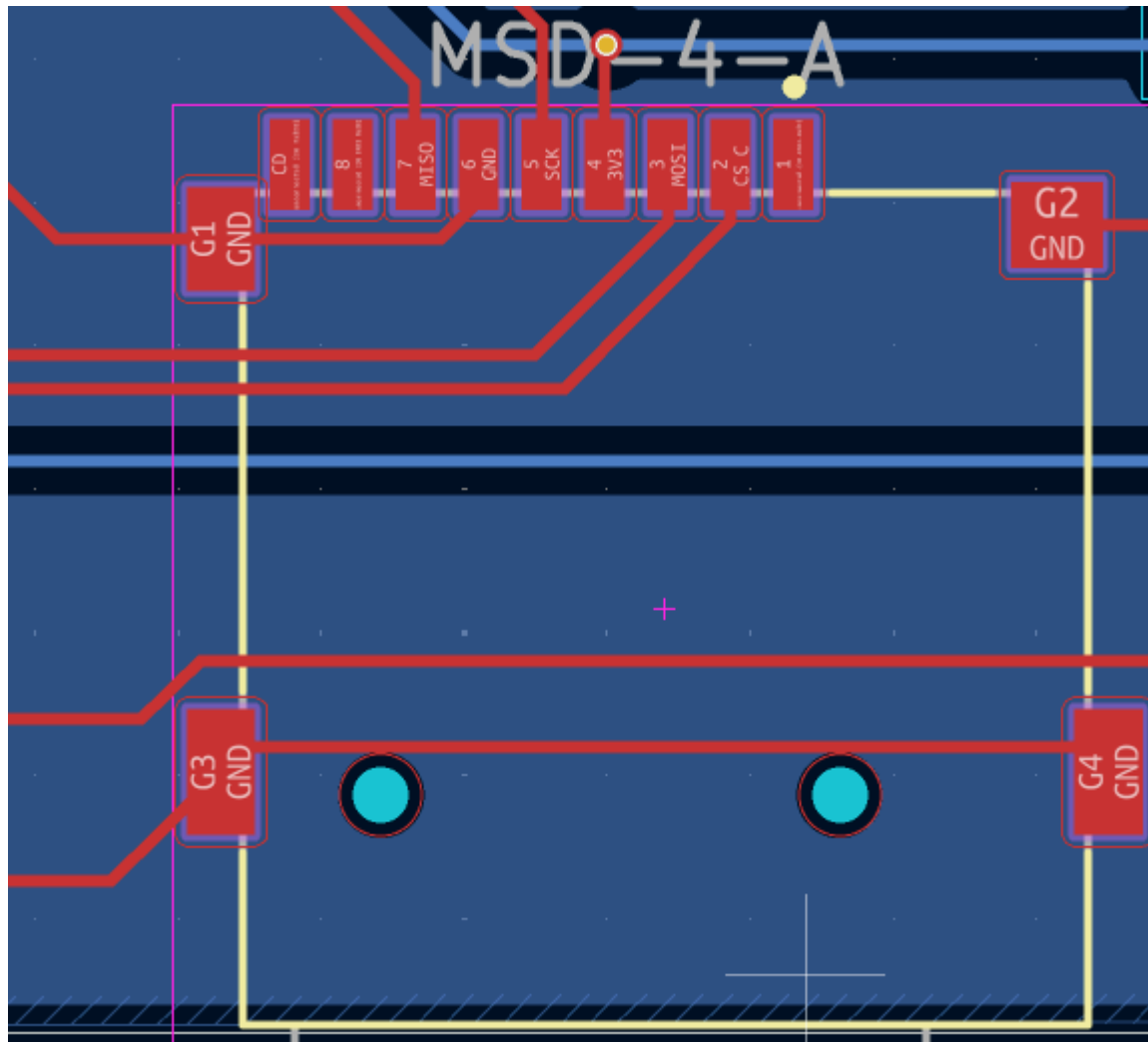
The IR led is connected to the ESP32 by the GPIO pin number 42, and the TSOP38238 is connected with the GPIO pin number 45.



Micro SD Slot

The micro SD slot was designed for a microSD card so with all the data that you have found you can store them in something much larger than the flash memory of the ESP32.

The micro SD slot is connected via SPI and the CSN pin is connected to the GPIO pin number 8.



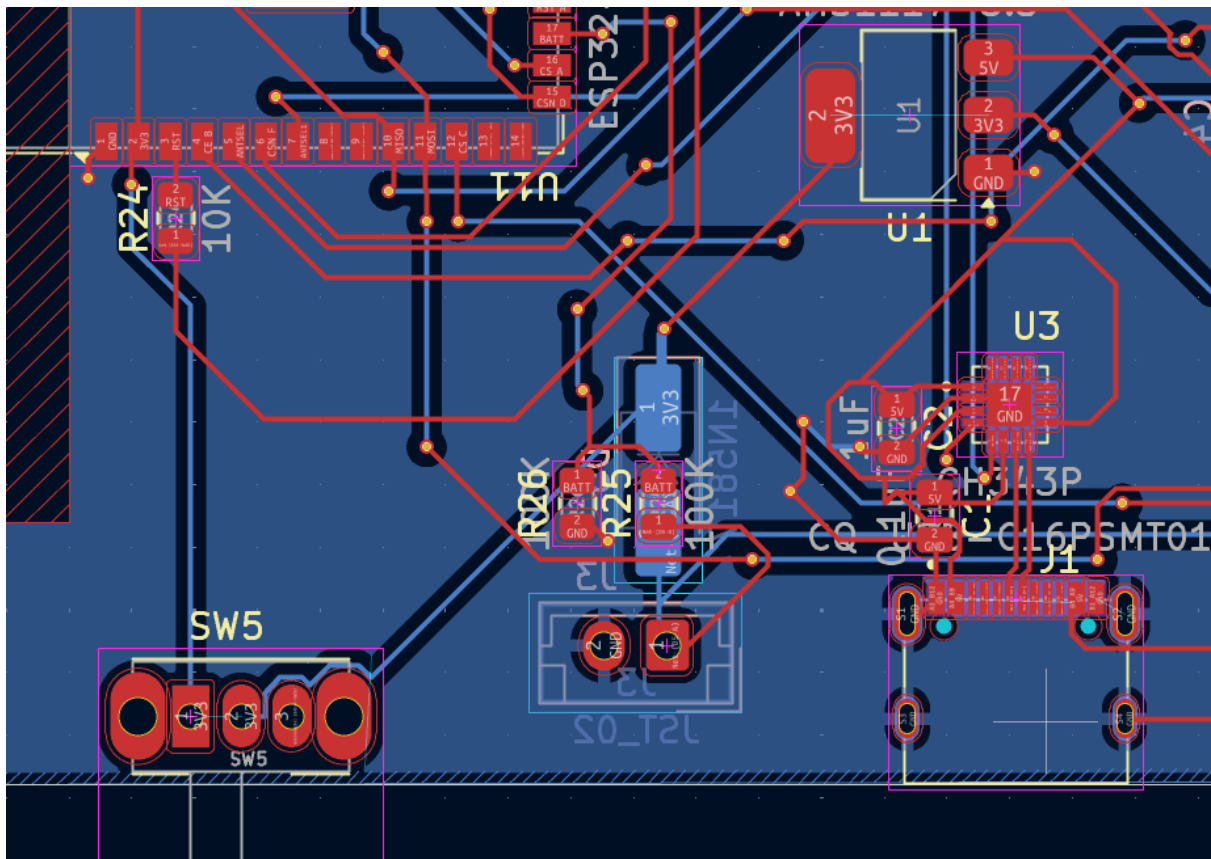
Power delivery system

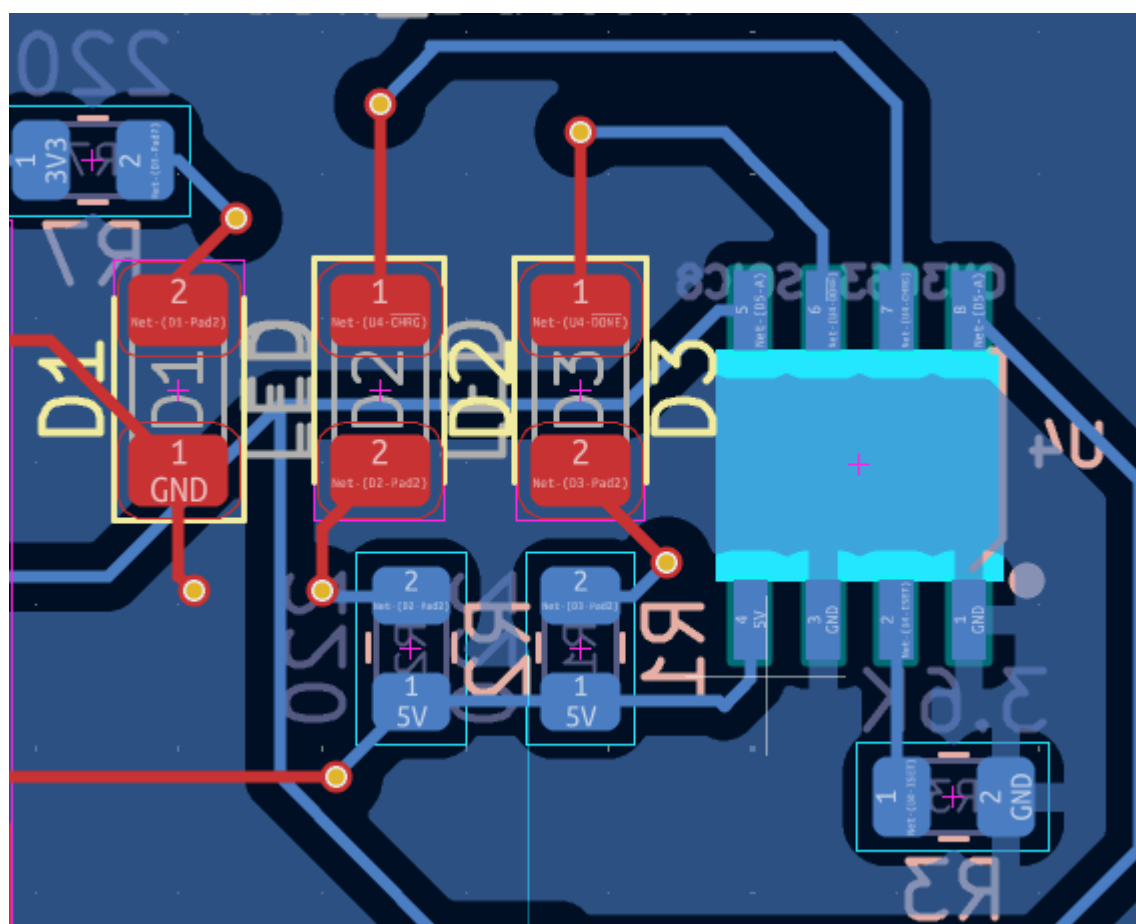
The power delivery and data delivery system is divided into 3 parts.

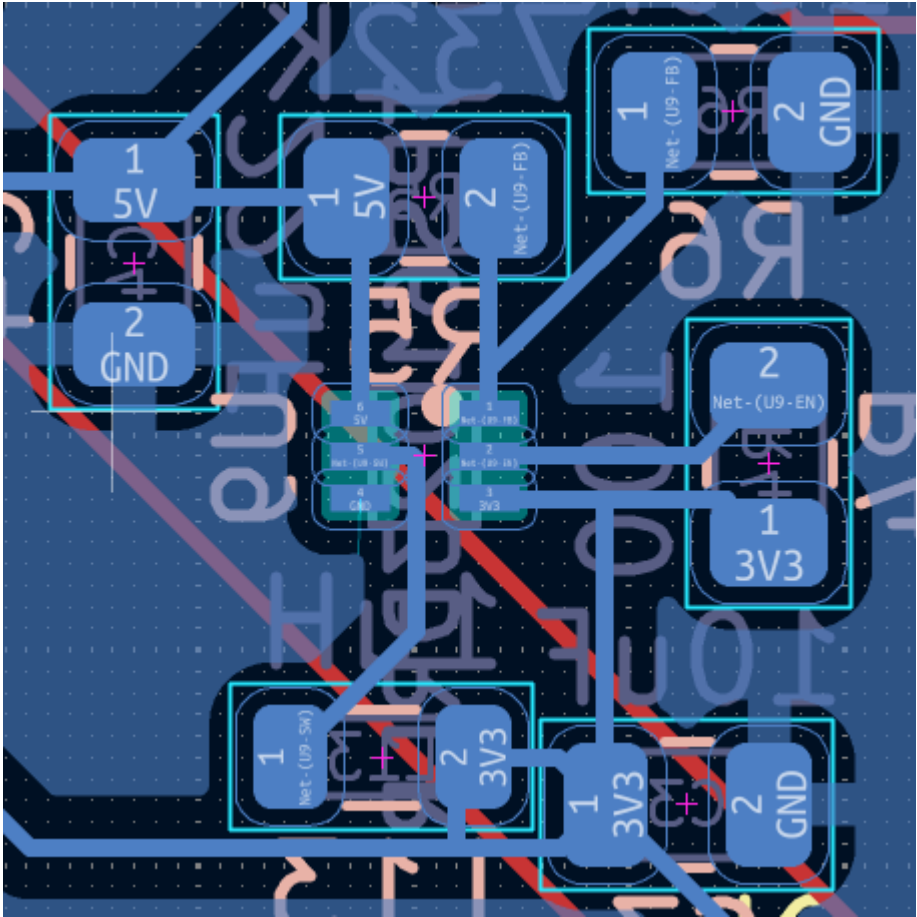
The first part is where the electricity enters, there are the USB-C port, the COM converter the CH343P, the ON/OFF switch, the battery connector, a diode so the current can enter the battery only by the battery charger, that is in the second part, 2 resistors that with some code can determine the charge of the battery, and last but not least the AMS1117-3.3 that converts the current from 5 volts into 3.3 volts.

In the second part there is the battery charger and protector of the CN3063 that help keep the battery healthy, and there are also the leds that tell you the state of the system, from active to charging to finished charging.

The third part has the TPS61023DRLR that converts the current from 3.3 volts into 5 volts when the Kraky2 is on battery mode, so there isn't any 5 volt current in the system.





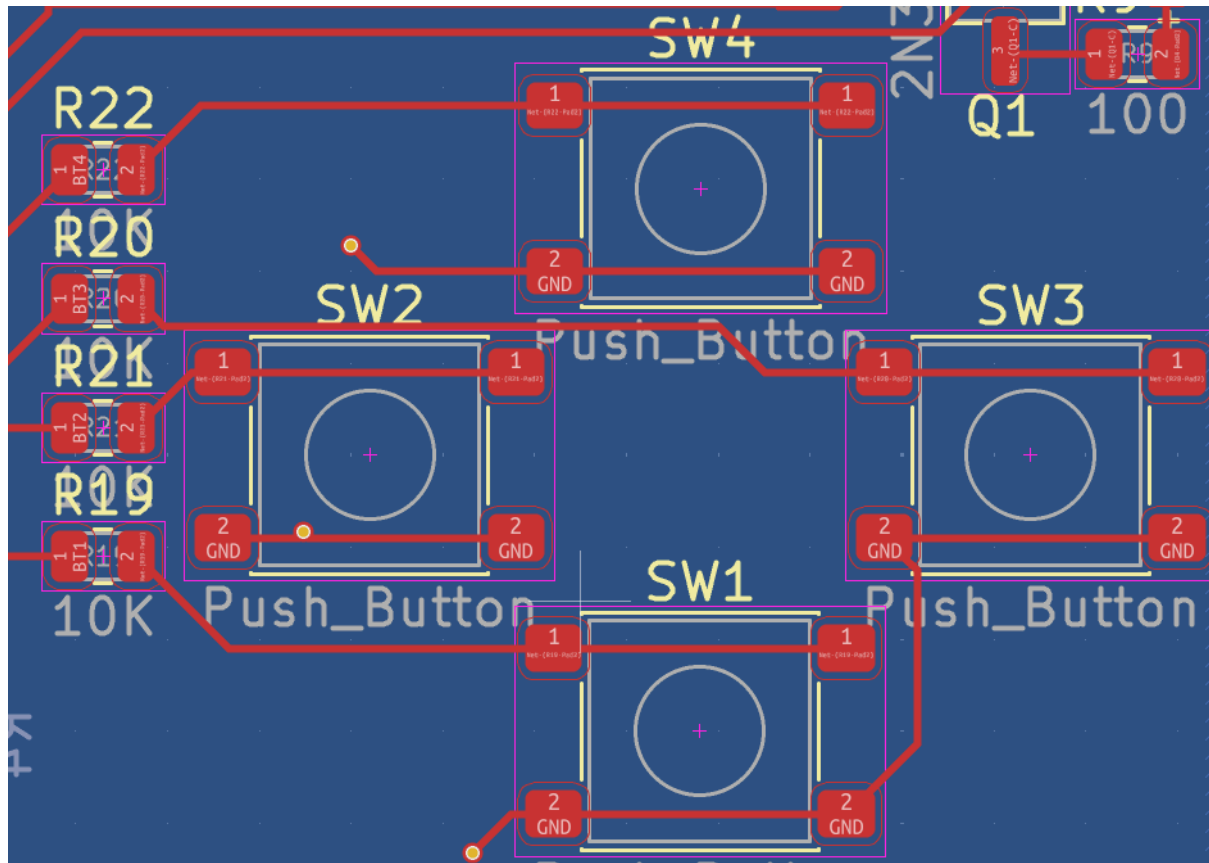


Navigation buttons

The buttons serve the function of navigating the firmware that we have created, they divide in UP, the switch 4, DOWN, the switch 1, OK, the switch 3, and BACK, the switch 2.

Each of the switches are connected to the ESP32, the switch 4 is connected to the GPIO pin number 41, the switch 1 is connected to the GPIO pin number 38, the switch 3 is connected to the GPIO pin number 40 and the switch 2 is connected to the GPIO pin number 39.

They are each connected to a 10K Ω pull-up resistor so the signal can be more clean.



The display

The display is a tft 1.8" 128*160 display, with the driver ST7735ST.

The display is connected to the ESP32 via SPI, and his CSN is the GPIO pin number 46, the DC is the GPIO pin number 47, and his RST is the GPIO pin number 10.

